Prob. 10. (Z&T. Ch. 8, 2.2)

Write down the formula sol-n for the IVP:

(1) $u_{xx} - \frac{1}{c^2} u_{tt} = 0 \qquad x \in \mathbb{R},\ t > 0$

(2) $u(x,0) = \varphi(x), \qquad u_t(x,0) = \psi(x), \qquad x \in \mathbb{R}.$

Did you change the var-s for the u_t in cond. too? Then answer the ↓ questions:

(a) Suppose that $\varphi = \psi = 0$ in some open interval of length l. Up to what time can you be sure that $u = 0$ at the center of the interval? Hint: you may assume $x = 0$ at the center of the interval [because using the hint in Pr. 9 part (a), we can determine the coordinate system <u>wherever</u> we <u>want</u>.]

(b) Suppose that $\varphi = \psi = 0$ outside of the interval $|x| \le 1$. Up to what time can you be sure that $u = 0$ at (i) $x = 3$, (ii) $x = 10$, & (iii) $x = -5$?

Sol-n: [Preamble]

By Pr. 9 we know the sol-n for $\boxed{u_{xx} - u_{tt} = 0}$, so let's use change of var-s to force (1) to look like this. For this, consider

(3) $\boxed{\tau = c \cdot t}$. Then, using the <u>chain rule</u>,

(4) $\frac{\partial u}{\partial t} = \frac{\partial u}{\partial \tau} \frac{d\tau}{dt} = c \cdot u_\tau$

(5) $\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial t}\left[\frac{\partial u}{\partial t}\right] = \frac{\partial}{\partial t}\left[c \cdot u_\tau\right] = c \frac{\partial u_\tau}{\partial \tau} \cdot \frac{d\tau}{dt} = c^2 \cdot u_{\tau\tau}.$

Then, using (5), (1) becomes

(6) $u_{xx} - \frac{1}{c^2} \cdot (c^2 \cdot u_{\tau\tau}) = u_{xx} - u_{\tau\tau} = 0$

And, using (4), (2) becomes, $t = c\tau \Rightarrow \boxed{\tau = \frac{t}{c}}$

$$u(x,t)\Big|_{t=0} = u(x,\tau)\Big|_{\tau=0} = u(x,0) = \varphi(x) \quad \&$$

$$u_t(x,t)\Big|_{t=0} = c\, u_\tau(x,\tau)\Big|_{\tau=0} = c\, u_\tau(x,0) = \psi(x), \quad \text{that is}$$

(7) $u(x,0) = \varphi(x), \qquad u_\tau(x,0) = \frac{1}{c}\cdot\psi(x).$

But we know the sol-n for (6) + (7):

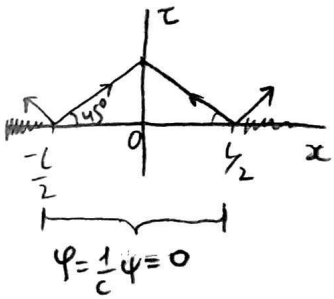
(8) $$u(x,\tau) = \frac{1}{2}\left[\varphi(x+\tau) + \varphi(x-t)\right] + \frac{1}{2}\int_{x-\tau}^{x+\tau} \frac{1}{c}\psi(s)\,ds,$$

replacing in (8), $\tau = \frac{t}{c}$, we get the sol-n for the original PDE:

(9) $$\boxed{u(x,t) = \frac{1}{2}\left(\varphi\left(x+\frac{t}{c}\right) + \varphi\left(x-\frac{t}{c}\right)\right] + \frac{1}{2c}\cdot\int_{x-\frac{t}{c}}^{x+\frac{t}{c}} \psi(s)\,ds}.$$

(a). To answer this question, we can answer first for $u(x,\tau)$ sol-n, where we know $u(x,\tau) = F(x-t) + g(x+t)$ is the general sol-n & $\xi_1 = \langle 1, 1\rangle$ & $\xi_2 = \langle 1, -1\rangle$. Then replace $\boxed{\tau = \frac{t}{c}}$ for the asked actual question:

(10)



$\Rightarrow$ By Pr. 9, (a),

$u(x,\tau) = 0$ until $\boxed{\tau = \frac{L}{2}}$. Then

$$\tau = \frac{t}{c} = \frac{L}{2} \Rightarrow \boxed{t = \frac{cL}{2}}$$

Hence $u(x,t) = 0$ until $\boxed{t = \frac{cL}{2}}$.

(b) Similarly like in part (a), let's answer $u(x,\tau)$ sol-n first then generalize it to $u(x,t)$ by $\tau = t/c$:

By Pr. 9, part (b) for the ↓ IVP that we obtained earlier,

(6) $u_{xx} - u_{\tau\tau} = 0 \qquad x \in \mathbb{R}, \quad t > 0, \quad c > 0$

(7) $u(x,0) = \varphi(x), \quad u_\tau(x,0) = \frac{1}{c} \cdot \psi(x), \quad x \in \mathbb{R}$

(**) Since $\varphi \neq 0$ & $\frac{\psi}{c} \neq 0$ on $|x| \leq 1$, these parameters, $\underbrace{\varphi \neq 0, \ \frac{\psi}{c} \neq 0,}$ can influence $u(x,\tau)$ until $\tau = 1$, meaning after that, they do NOT have any influence on $u(x,\tau)$ to make $u(x,\tau)$ nonzero. Hence if $t > 1$, then $u(x,\tau) = 0$. That is

(8)
(i) $x = 3, \quad u(x,\tau) = 0$ for $\tau > 1$
(ii) $x = 10, \quad u(x,\tau) = 0$ for $\tau > 1$
(iii) $x = -5, \quad u(x,\tau) = 0$ for $\tau > 1$

Converting these in (8) into $u(x,t)$ with $\tau = \frac{t}{c}$:

(i) $x = 3, \quad u(x,t) = 0$ for t; $\frac{t}{c} > 1 \Rightarrow t > c$.

(ii) $x = 10, \quad u(x,t) = 0$ for all t, $\frac{t}{c} > 1 \Rightarrow t > c$.

(iii) $x = -5, \quad u(x,t) = 0$ for all t, $\frac{t}{c} > 1 \Rightarrow t > c$,

answering the asked questions. ∎